Syntactic height impacts prosodic size: An argument for cyclic prosodification

Paula Fenger, Nadja Fiebig, Sören Tebay & Philipp Weisser

18 October 2024

Introduction

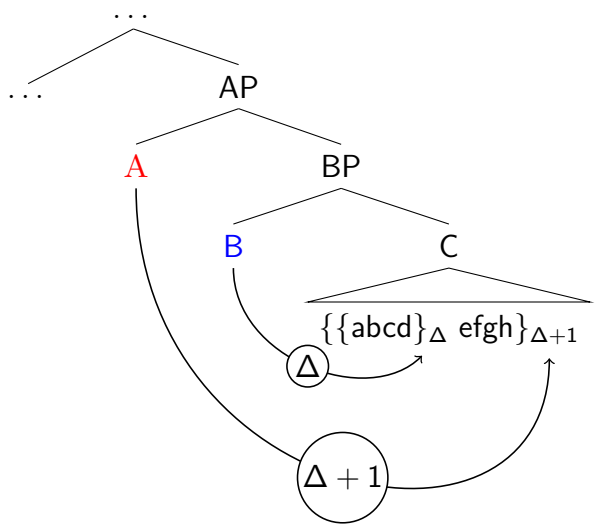
- We provide two crosslinguistically robust generalizations concerning the Morphosyntax-Phonology Interface coming from independent phenomena:
  - Prosodically determined second position coordinators
  - Verbal Number Reduplication
- The two generalizations converge on the same abstract correlation:

- We provide two crosslinguistically robust generalizations concerning the Morphosyntax-Phonology Interface coming from independent phenomena:
  - Prosodically determined second position coordinators
  - Verbal Number Reduplication
- The two generalizations converge on the same abstract correlation:

Correlation of Syntactic Height and Prosodic Size (COSHAPS):

If two syntactic heads (or their respective exponents) trigger the same process on an element X, then the process of the syntactically higher head will apply to a bigger (or equal) prosodic domain within X.

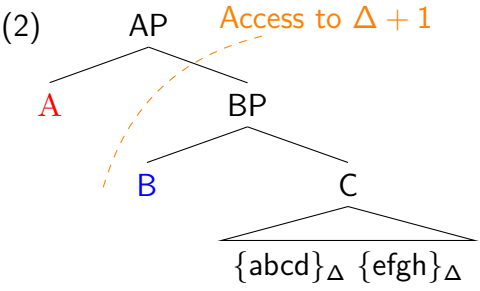
(1) The Correlation of Syntactic Height and Prosodic Size:



- We claim that this provides a strong argument for a cyclic account of the prosodification of syntactic structure.

- We claim that this provides a strong argument for a cyclic account of the prosodification of syntactic structure.
- At the point where B is inserted and triggers the application of a phonological process, only the lower prosodic domain Δ is built
  - ↪ The higher domain $\Delta + 1$ is not accessible yet as it is only completed after B.

(2)



- The crucial piece of information only becomes accessible at a later stage in the derivation.

Take-home message:

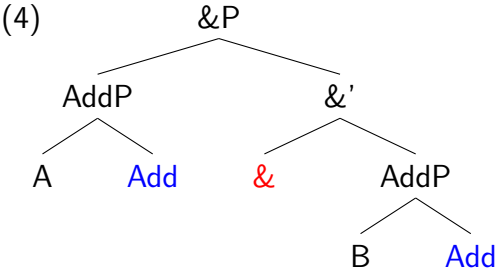
- The COSHAPS indicates that smaller domains must be construed and become accessible before bigger domains.
- This finding is at odds with parallel computation of prosodic domains (Truckenbrodt 1999; Selkirk 2011; Elfner 2015 and many others).

Shifting Coordinators

- Languages typically use two morphological strategies to coordinate elements.

(3)
a. Petyr-en Maša-jen
Peter-ADD Masha-ADD
b. Petyr no Maša
Peter and Masha
'Peter and Masha'
c. Petyr-en no Maša-jen
Peter-ADD and Masha-ADD
'Peter and Masha'
Udmurt, (Weisser, 2017)

(4)



e.g. Bhatt (2014),
Mitrović and Sauerland (2014)

- Some clause-level coordinators do not appear in their canonical position (e.g. in between the two clauses) but shift to a second position inside one of their coordinands.
  - ↪ And in some of the cases, this position is prosodically defined:

(5) a. ... sine=**que** ferr-o fu-erint
without=COORD iron-ABL be-SBJ.PERF.3PL
'... and that they were without swords' Latin, Embick (2007, 309)

b. ... **que** $\{\text{sine}\}_\omega$ $\{\text{ferro}\}_\omega$ $\{\text{fuerint}\}_\omega$

(6) a. ... Ọlá ò bá **sì** lọ.
... Ola should have COORD go
'...and Ola should have gone.' Yorùbá, Aremu and Weisser (2024)

b. ... **sì** $\{\text{Ọlá ò bá}\}_\phi$ $\{\text{lọ}\}_\phi$

- Weisser (2024): Cross-linguistic study of non-canonical coordinator placement (48 languages/36 genera/21 families).

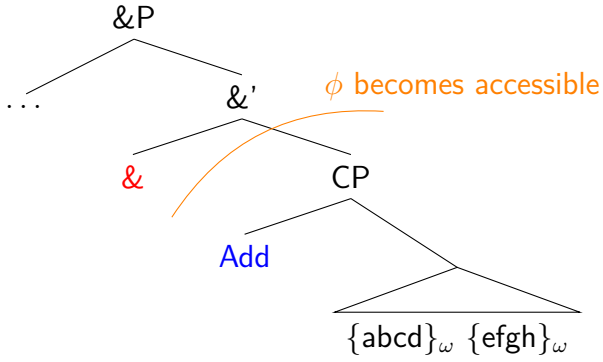
	[ω-COORD	[ϕ-COORD
ADD	Latin Cherokee Kalaallisut Anc. Greek Hittite Gothic Old Irish	–
&	–	Yorùbá German Polish

Generalization

When syntactic ADD-coordinators shift onto a prosodic constituent, they move to the first phonological word (ω); &-coordinators move after the first phonological phrase (ϕ).

- We observe a correlation between the syntactic height of the coordinator and the size of the prosodic constituent that is affected by it.
- Only once the CP is built, the ϕ-phrases within it become an accessible target.

(7)



Flavours of verbal number

- The morphosyntactic VERBAL NUMBER comes in two flavours:
  - Argument Number: tracks singular/plural of an argument (participant #)
  - Event Number: tracks single/multiple events (pluractionality)

(8) Kipsigis (Kouneli, 2020, 7)

a. labat-i là:kwɛ̀:t.
run.**sg**-IPFV child.NOM
'The child is running.' (baseline)

b. ruaj là:gô:k.
run.**pl**.IPFV children.NOM
'The children are running.' (A#)

- The morphosyntactic VERBAL NUMBER comes in two flavours:
  - Argument Number: tracks singular/plural of an argument (participant #)
  - Event Number: tracks single/multiple events (pluractionality)

(8) Kipsigis (Kouneli, 2020, 7)

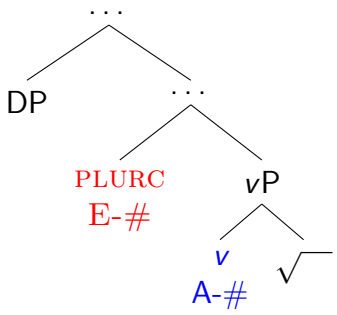
a. labat-i là:kwɛ̀:t.
run.**sg**-IPFV child.NOM
'The child is running.' (baseline)

b. ruaj là:gô:k.
run.**pl**.IPFV children.NOM
'The children are running.' (A#)

c. ruaj∼ruaj là:gô:k.
run.**pl**∼**red** children.NOM
'The childen are running **over and over again**.' (E+A#)

- Argument number is very close to the verb root: *v* or lower
  - A-# can pick out # from closest DP
  - ↪ # ends up on *v* through Agree with higher head
- Event number is higher: outside *vP* on PLURC (pluractionality).

(9)



e.g.,Thornton (2017, 2019, 2020); Amato (2018), Fiebig (2024)

- Both types are frequently expressed by means of reduplication.

(10) Niuean, Haji-Abdolhosseini et al. (2002)

a. kua ha-hala e ia e tau lā akau.
PERF RED.ARG-cut ERG he ABS PL branch tree
'He cut the branches.' (A#)

b. ne noko-noko e ia e gutuhala
PST RED.EVENT-knock ABS she ABS door
'She knocked on the door (many times).' (E#)

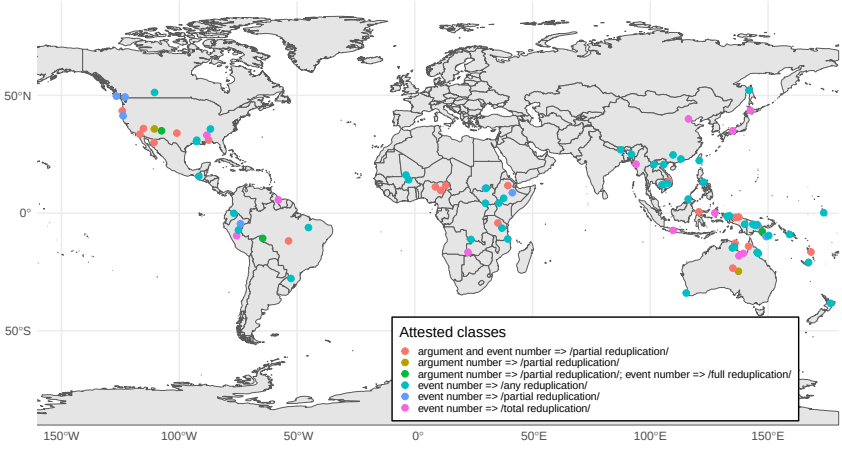
- Both types are frequently expressed by means of reduplication.

(10) Niuean, Haji-Abdolhosseini et al. (2002)

a. kua ha-hala e ia e tau lā akau.
PERF RED.ARG-cut ERG he ABS PL branch tree
'He cut the branches.' (A#)

b. ne noko-noko e ia e gutuhala
PST RED.EVENT-knock ABS she ABS door
'She knocked on the door (many times).' (E#)

- Crucially, we find a correlation between the size of the reduplicant and the type of verbal number


50°N
0°
50°S
150°W
100°W
50°W
0°
50°E
100°E
150°E
Attested classes
argument and event number => /partial reduplication/
argument number => /partial reduplication/
argument number => /partial reduplication/; event number => /full reduplication/
event number => /any reduplication/
event number => /partial reduplication/
event number => /total reduplication/

- Fiebig (2024): Typological investigation of reduplication of verbal number: 97 languages from 37 families.
  - Partial reduplication marks both flavours of verbal number, but full reduplication marks only event number. (cf. Bybee et al. 1994)

class	partial reduplication	full reduplication	# of L
1	-	event-#	13
2	event-#	-	7
3	argument-#	-	2
4	argument-#	event-#	4
5	argument-# / event-#	-	20

- Fiebig (2024): Typological investigation of reduplication of verbal number: 97 languages from 37 families.
  - Partial reduplication marks both flavours of verbal number, but full reduplication marks only event number. (cf. Bybee et al. 1994)

class	partial reduplication	full reduplication	# of L
1	-	event-#	13
2	event-#	-	7
3	argument-#	-	2
4	argument-#	event-#	4
5	argument-# / event-#	-	20
6	argument-# / event-#	argument-#	Unattested

Generalization

Partial reduplication expresses event- and argument number, but full reduplication only ever marks event number.

Generalization

Partial reduplication expresses event- and argument number, but full reduplication only ever marks event number.

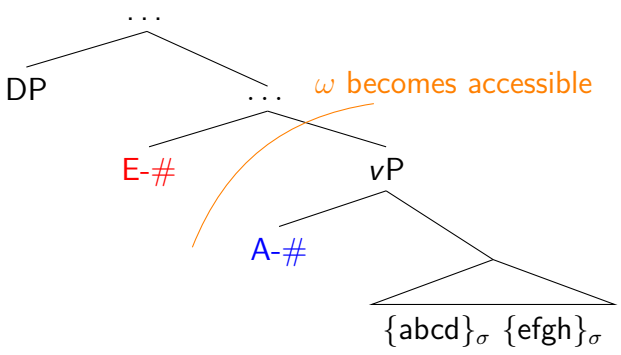
⇒ Full reduplication never marks argument number.

	partial reduplication	full reduplication
Argument number	✓	✗
Event number	✓	✓

- The lower head will target a prosodic constituent below the ω-level (e.g. a syllable).
- The higher head can target both the ω-level and or a smaller constituent.

- The lower head will target a prosodic constituent below the ω-level (e.g. a syllable).
- The higher head can target both the ω-level and or a smaller constituent.
- This can be made sense of by assuming that the crucial ω is only accessible after the *v*P is completed.

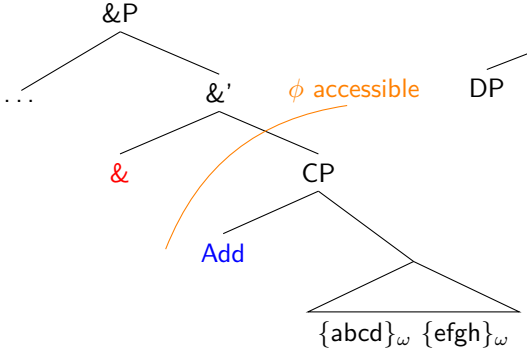
(11)



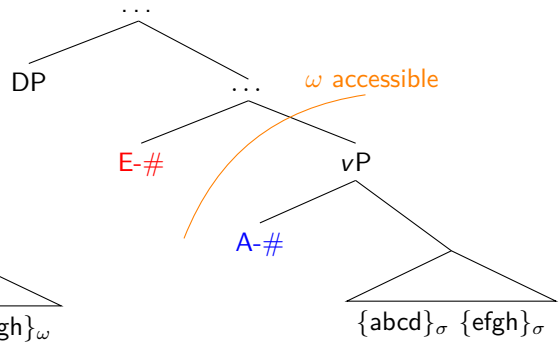
Deriving the patterns

An abstract similarity between two different phenomena:

(12)



(13)

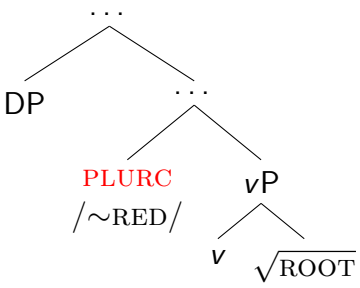


- Syntactic domains delimit which prosodic structure is built (Kratzer and Selkirk, 2007; Pak, 2008; Newell, 2008; Kahnemuyipour, 2009)
- Prosodic structure is built incrementally (Inkelas, 1990; Bermudez-Otero, 2019)
- VI occurs bottom-up (Bobaljik, 2000)
- When VI of a node takes place, the exponent is linearized wrt. to previously generated structure (Kalin, 2022; Weisser, 2024)
- Operations such as Reduplication (COPY) or clitic placement (INTEGRATE) are sensitive to the prosodic structure built at that point

Full Reduplication:

1. E-# is on PLURC, above *v*P (14)
2. *v*P phase to spell-out, bottom up VI & linearization:
   1. All heads in *v*P into ω
3. Higher phase: PLURC is linearized wrt ω, (15)

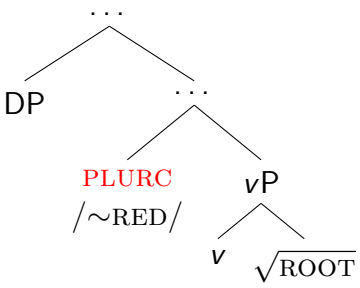
(14)



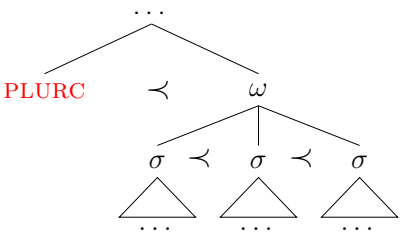
Full Reduplication:

1. E-# is on PLURC, above *v*P (14)
2. *v*P phase to spell-out, bottom up VI & linearization:
   1. All heads in *v*P into ω
3. Higher phase: PLURC is linearized wrt ω, (15)

(14)



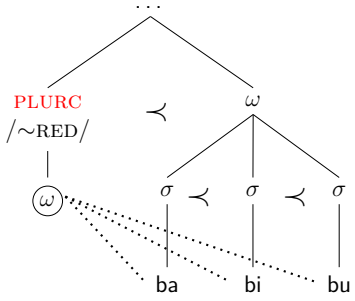
(15)



Full Reduplication:

- RED, on PLURC is spelled out (16). ⇒ ω is the target

(16)



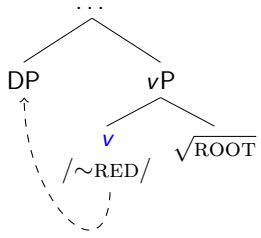
⇒ /babibu∼babibu/

(17) REDUPLICATION
Fill the prosodic constituent of a segmentally empty vocabulary item with material from the adjacent prosodic unit of the same size. (based on Marantz 1982)

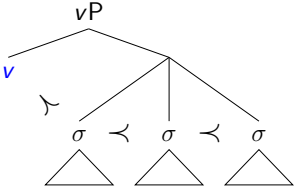
Partial Reduplication:

- *v* Agrees with a higher head (PLURC or DP) (Thornton, 2018; Amato 2018)
- *v*P phase to spell-out, bottom up VI
  1. $\sqrt{root}$ is not prosodified (only has σ)
  2. *v* is linearized wrt structure built

(18)



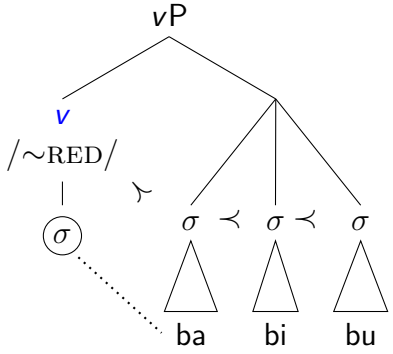
(19)



Partial Reduplication:

- RED is spelled out on *v* (20) ⇒ there is no ω, only σ is visible to copy:

(20)



⇒ /ba∼babibu/

- The basic building blocks (e.g. Vocabulary Insertion, Prosodification, Reduplication) all work the same in both configurations.
- The difference in Reduplication of the two types of syntactic elements arises purely as the result of the different timing.
  - ↪ It is crucial that prosodification applies cyclically.
  - ↪ As soon as a given cyclic node in the tree is reached (e.g. *v*P), it will be mapped to the respective prosodic category.

- The basic building blocks (e.g. Vocabulary Insertion, Prosodification, Reduplication) all work the same in both configurations.
- The difference in Reduplication of the two types of syntactic elements arises purely as the result of the different timing.
  - ↪ It is crucial that prosodification applies cyclically.
  - ↪ As soon as a given cyclic node in the tree is reached (e.g. *v*P), it will be mapped to the respective prosodic category.
- This is the same for the shifting coordinators:
  - ↪ The syntactic domains are inside (ADD) or outside (&) CP
  - ↪ Only when CPs are built, bigger phonological domains are visible (ι)
  - ↪ The shifting operation (INTEGRATION in Weisser 2024) is sensitive to the amount of structure being built

Conclusion

- We reported two case studies of robust crosslinguistic generalizations about the respective subtypes of a morphosyntactic category and the size of the prosodic domain they affect if they trigger a phonological process.

(21) **Coordination**

	[ω-COORD	[ϕ-COORD
ADD	✓	✗
&	✗	✓

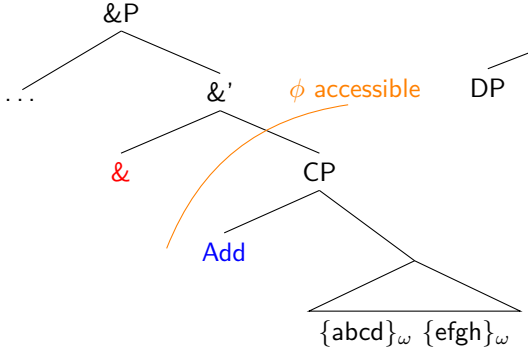
(22) **Verbal Number**

	Part.RED	full-RED
A-#	✓	✗
E-#	✓	✓

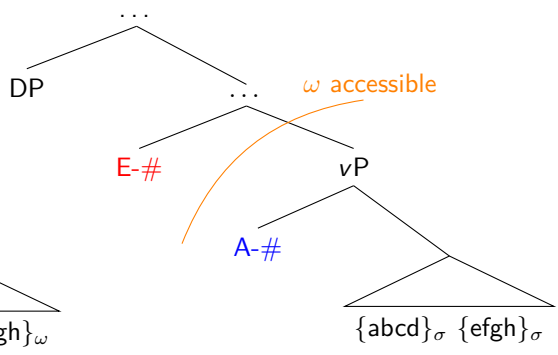
Correlation of Syntactic Height and Prosodic Size (COSHAPS):

If two syntactic heads (or their respective exponents) trigger the same process on an element X, then the process of the syntactically higher head will apply to a bigger (or equal) prosodic domain within X.

(23)



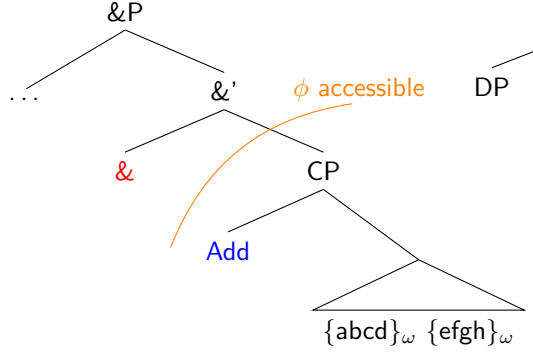
(24)



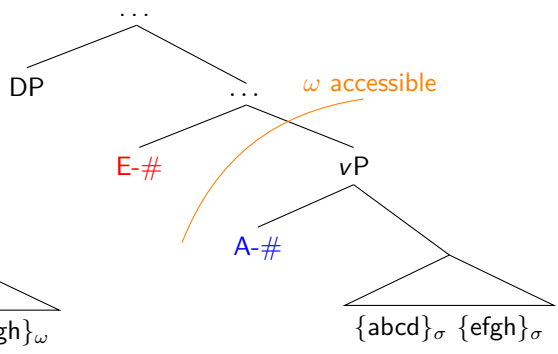
Correlation of Syntactic Height and Prosodic Size (COSHAPS):

If two syntactic heads (or their respective exponents) trigger the same process on an element X, then the process of the syntactically higher head will apply to a bigger (or equal) prosodic domain within X.

(23)



(24)



⇒the COSHAPS presents a strong argument for a cyclic conception of prosodification.

Literature I

Amato, I. (2018). A morphosyntactic account to verbal number in Mupun. In *Proceedings of the 26th conference of the student organization of linguistics in Europe*, pages 20–44.

Aremu, D. and Weisser, P. (2024). Prosodically determined coordinator placement in yorùbá. *Glossa: A journal of general linguistics*.

Bermudez-Otero, R. (2019). Cyclicity. In *The Blackwell companion to phonology*. Wiley Blackwell.

Bhatt, R. (2014). Coordination and finiteness in malayalam: A commentary on jayaseelan (2013). *Natural Language & Linguistic Theory*, 32(1):213–229. Special Issue: Finiteness in South Asian Language.

Bobaljik, J. (2000). The ins and outs of contextual allomorphy. In Grohmann, K. and Struijke, C., editors, *University of Maryland Working Papers in Linguistics*, pages 35–71.

Bybee, J. L., Perkins, R., and Pagliuca, W. (1994). *The evolution of grammar: Tense, aspect and modality in the languages of the world*. The University of Chicago Press, Chicago.

Literature II

Elfner, E. (2015). Recursion in prosodic phrasing: Evidence from connemara irish. *Natural Language and Linguistic Theory*, 33:1169–1208.

Embick, D. (2007). Linearization and local dislocations, derivational mechanics and interactions. *Linguistic Analysis*, 33(303–336).

Haji-Abdolhosseini, M., Massam, D., and Oda, K. (2002). Number and Events: Verbal Reduplication in Niuean. *Oceanic Linguistics*, 41(2):475–492.

Inkelas, S. (1990). *Prosodic Consituency in the Lexicon*. New York: Garland.

Kahnemuyipour, A. (2009). *The Syntax of Sentential Stress*. Oxford University Press.

Kalin, L. (2022). Infixes really are (underlyingly) prefixes/suffixes: Evidence from allomorphy on the fine timing of infixation. *Language*, 98(4):641–682.

Kouneli, M. (2020). Layers of morphosyntactic number in Eastern Sudanic. DFG grant proposal.

Literature III

Kratzer, A. and Selkirk, E. (2007). Phase theory and prosodic spellout: The case of verbs. *The Linguistic Review*, 24:93–135.

Marantz, A. (1982). Re reduplication. *Linguistic Inquiry*, 13(3):435–482.

Mattiola, S. (2019). *Typology of pluractional constructions in the languages of the world*. Typological studies in language. John Benjamins Publishing Company, Amsterdam.

Mitrović, M. and Sauerland, U. (2014). Decomposing coordination. In Iyer, J. and Kusmer, L., editors, *Proceedings of the NELS 44*, pages 39–52. Amherst: GLSA.

Newell, H. (2008). *Aspects of the morphology and phonology of phases*. PhD thesis, McGill University.

Pak, M. (2008). *The postsyntactic derivation and its phonological reflexes*. PhD thesis, University of Pennsylvania.

Selkirk, E. (2011). The syntax-phonology interface. In Goldsmith, J., Riggle, J., and Yu, A., editors, *The handbook of phonological theory, 2nd edition*. Wiley Blackwell.

Literature IV

Thornton, A. (2017). Reduplicating verbal plurals cross-linguistically. In *Proceedings of the 43rd Annual Meeting of the Berkeley Linguistics Society*.

Thornton, A. (2019). Agreeing in number: Verbal plural suppletion and reduplication. *The Linguistic Review*, 36:531–552.

Thornton, A. (2020). *Morphophonological and Morphosyntactic Domains*. Doctoral dissertation, University of Connecticut.

Truckenbrodt, H. (1999). On the relation between syntactic phrases and phonological phrases. *Linguistic Inquiry*, 30:219–56.

Weisser, P. (2017). Three types of Coordination in Udmurt. In Kudryavtseva, R. A., editor, *Problems in Mari and Comparative Philologies. Yoshkar-Ola*.

Weisser, P. (2024). A typology of shifted coordinators and what they tell us about clitics and their formal modelling. Habilitation Thesis: Universität Leipzig.

Wiltschko, M. (2008). The syntax of non-inflectional plural marking. *Natural Language & Linguistic Theory*, 26:639–694.

Literature V

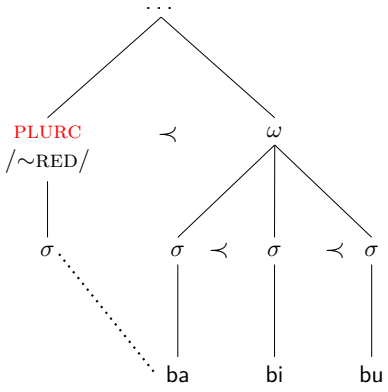
Wiltschko, M. and Steriopolo, O. (2007). parameters of variation in the syntax of diminutives. *Proceedings of the 2007 annual conference of the Canadian Linguistic Association*.

Wood, E. J. (2007). *The semantic typology of pluractionality*. Doctoral dissertation, University of California, Berkeley.

Appendix

option 1:
The PLURC head is spelled out by a skeleton of the size of a smaller prosodic unit (e.g. a σ) and can associate with this unit within the prosodic word.

(25)

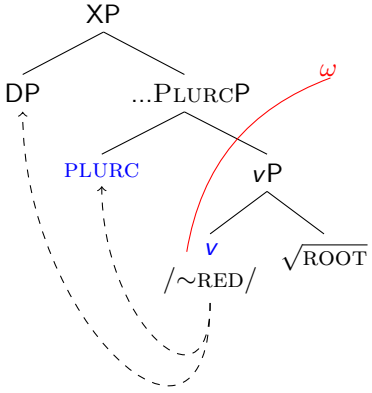


⇒ /ba∼babibu/

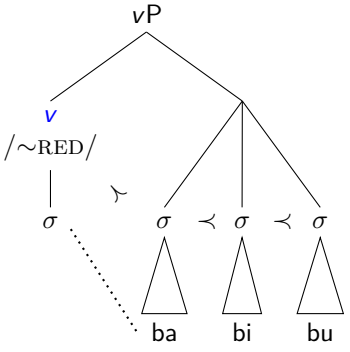
option 2:

PLURC is a head in some and an adjunct in other languages (similar to e.g. diminutives Wiltschko and Steriopolo 2007, Wiltschko 2008). The head version of PLURC would then be a possible goal for agreement with the probe (v) and would be spelled out just as low as the result of the agreement operation with a DP.

(26)



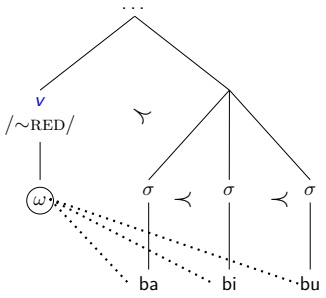
(27)



⇒ /ba∼babibu/

We argued that Argument number is never spelled out by total reduplication, as *v* cannot be spelled out by a skeleton of the size of ω.
But couldn't phonology just create such a ω in this configuration?

(28)



⇒ /babibu/

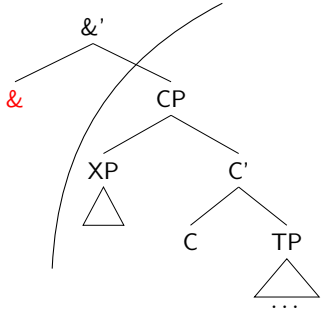
- The empty ω can associate with the segments which are not yet associated to another ω.
- Therefore, no reduplication takes place at all, since this integration is universally cheaper than all other options (besides deleting the floating ω).
  ⇒ The input equals the output and hence we cannot observe it.
- In the cases before, where total reduplication **does** take place, the syllables are dominated by a ω and therefore the integration cannot take place
  ⇒ reduplication takes place, basically, as a form of repair.

- In Weisser (2024), an approach is sketched that derives the pattern in a cyclic framework, linearization statements as well as prosodification are generated as part of a bottom-up Vocabulary Insertion process.
  - ↪ As soon as VI hits a syntactic node in the structure, it will map it into a prosodic constituent and linearize it wrt. the previously generated structure. (cf. Kalin 2022)
- Second position placement arises as a subcategorization-driven transformation (INTEGRATION) that dislocates an element X to a position after the first element within the category it precedes.

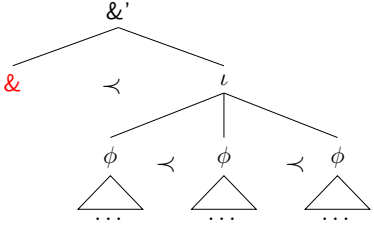
Phonological 2nd positions - A monosyndetic coordinator

- The configuration at Spell-Out (29) the configuration at Vocabulary Insertion (30)

(29)



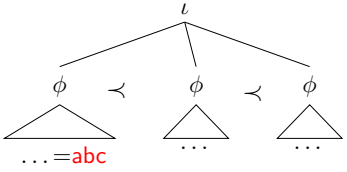
(30)



(31) [&] ⟷ /-abc/ [⇨⇦]

The result of INTEGRATION:

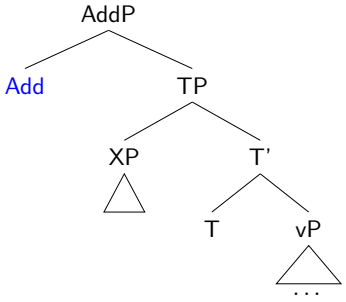
(32)



Phonological 2nd positions - A polysyndetic coordinator

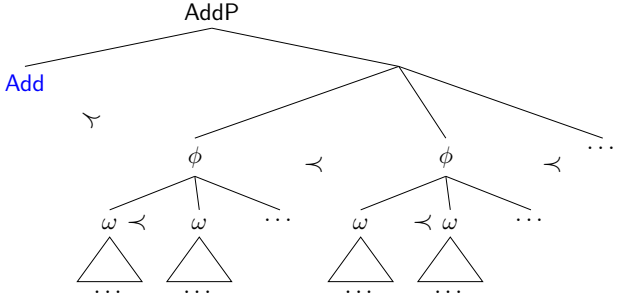
- The configuration at Spell-Out (33)

(33)



- The configuration at Vocabulary Insertion:

(34)



(35) [Add] $\longleftrightarrow$ /-abc/ [⇨⇦]

The result after INTEGRATION

(36)

